Thales A. Amaral

AI & Data Engineer

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Summary

Software engineer (R&D) at SPEED Lab (DCC/UFMG) and Computer Science student at UFMG. Experience with AI agents, LLMs (RAG, MCP, tool calling), data pipelines and cloud observability on Azure.

Experience

Software Engineer (R&D) — **Nov 2026 – Present**
SPEED Lab – Department of Computer Science (DCC), UFMG — Belo Horizonte, Brazil

- Software research and development funded by an EMBRAPII grant.

AI & Data Engineering Intern — **Oct 2025 – Oct 2026**
Rehagro — Belo Horizonte, Brazil

- Automated the **transcription and classification of 1,000+ educational videos** with a **Python, OpenAI Whisper and Apache Airflow** pipeline integrated with internal **PostgreSQL** databases.
- Set up **monitoring for about 80 Azure virtual machines** with **Grafana and Prometheus**, supporting usage, performance and infrastructure cost analysis.
- Made **VM, cost and network data queryable in natural language** through an **LLM agent with tool calling**.
- Built a **metadata catalog** for the company's main databases, documenting business rules, granularity and cardinality, in a **React/FastAPI app integrated with an AI agent**.
- Delivered solutions with **RAG, MCP and model evaluation**, from requirements with business teams to validation.
- Integrated **AI agents, automations and REST APIs** into the internal data platform using **Laravel and React**.
- Previously: Software Engineering Trainee at Rehagro (Jun – Sep 2025).

IT Support Technician — **Oct 2023 – Sep 2024**
Department of Toxicology – Federal University of São João del-Rei (UFSJ) — Divinópolis, Brazil

- Maintained the **IT infrastructure for researchers and laboratories**, solving software, hardware and network issues.

Education

B.Sc. in Computer Science — **Mar 2025 – 2029 (expected)**
Federal University of Minas Gerais (UFMG) — Belo Horizonte, Brazil

- Coursework: Data Structures, Algorithms, Databases, Discrete Mathematics and Probability.

Computer Technician, integrated with high school — **Mar 2022 – Dec 2024**
Federal Center for Technological Education of Minas Gerais (CEFET-MG) — Divinópolis, Brazil

- Software development, databases, networking and computer systems. Class representative in 2022.

Research and Projects

Junior Researcher – Drug Seizure Data Analysis — **Jan 2023 – Dec 2024**
CEFET-MG, with UFSJ and the Minas Gerais Military Police (PMMG) — Divinópolis, Brazil

- Structured and analyzed **drug seizure, prison system and geospatial data** in **Python**, resulting in a **published paper**. **FAPEMIG** undergraduate research grant (Oct 2023 – Sep 2024).

FruitAdvisor – Capstone Project — **2024**
CEFET-MG — Divinópolis, Brazil

- Mobile app that **recognizes fruits from a photo** using **Flutter** and a **convolutional neural network (CNN)**, showing nutritional information.

Skills

AI & Data: LLMs, RAG, MCP, Tool Calling, Model Evaluation, Machine Learning, OpenAI API, Whisper, Airflow, PostgreSQL, MySQL
Backend & Web: FastAPI, React, Node.js, Laravel, REST APIs
Infrastructure & Tools: Azure, Docker, Grafana, Prometheus, Git, Linux, Shell
Programming Languages: Python, C, C++, Java, JavaScript, Dart, SQL, HTML, CSS

Languages, Courses and Awards

Machine Learning Specialization – DeepLearning.AI and Stanford Online (Coursera), Oct 2026 – in progress
English C1 (full professional proficiency) – Cambridge English Certificate, score **181/190**; Portuguese (native)
Gold Medal – Brazilian National Science Olympiad (ONC), 2022